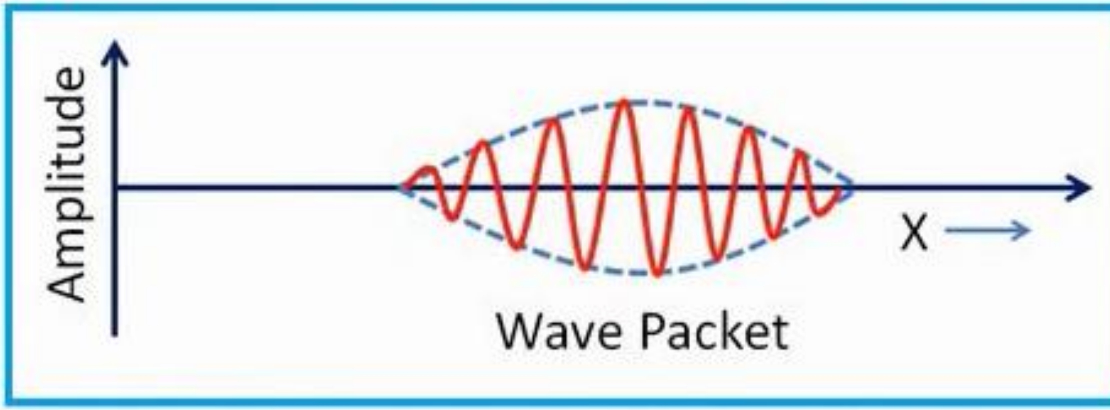
Unit-3
Lecture-4

Schrodinger's solution

- Group of waves
- Wave packet



✓ The phases and amplitude of the waves in group are such that they undergo constructive interference only over a small region where the particle may be located.

✓ Outside this region, destructive interference occurs and amplitude is zero.

✓ Velocity with which this wave packet moves is called as group velocity. This group velocity is equal to the velocity of particle.

The wave function Ψ

- Schrödinger assumed that a quantity ψ represents a De Broglie wave.
- It is called as a wave function.
- This wave function mathematically describes the motion of particle.
- It is not possible to locate the particle precisely at point (x,y,z).
- This wave function gives a probability of finding the particle at (x,y,z) at time t.

The wave function Ψ

- If particle exists, probability of finding the particle somewhere in the space must be unity.

$$\boxed{\iiint_{-\infty}^{\infty} \Psi^{*}\Psi \, dx \, dy \, dz = 1}$$

- This condition is called as normalization condition. The wave function is normalized. It means, it satisfies this condition.

The wave function Ψ must fulfill the following conditions -

- ψ must be finite, continuous and single valued everywhere.
- Its derivative, $\frac{\partial \Psi}{\partial x}$ or $\frac{\partial \Psi}{\partial t}$ must also be finite, continuous and single valued everywhere.
- ψ must have atleast some physically acceptable solutions.
- ψ must obey the principle of linear superposition i.e. ψ can anytime be expressed as a linear combination of two wave functions say

$$\Psi(x, t) = A\phi_1(x, t) + B\phi_2(x, t)$$

Schrodinger's Time Independent Wave Equation (STIE)

According to the De Broglie theory, a particle of mass 'm' moving with velocity 'v' is associated with a wave of wavelength-

$$\boxed{\lambda = \frac{h}{mv}} \quad (1)$$

The wave equation of stationary wave associated with the particle in terms of Cartesian co-ordinate system at any instant is given by – $\Psi = \Psi_0 \sin\omega t$

$$\boxed{\therefore \Psi = \Psi_0 \sin 2\pi \nu t} \quad (2)$$

Where ψ_0 is the amplitude at the point under consideration

One dimensional classical differential equation of wave motion can be written as –

$$\boxed{\frac{\partial^2 \Psi}{\partial t^2} = v^2 \frac{\partial^2 \Psi}{\partial x^2}} \quad (3)$$

Differentiating equation (2)

$$\frac{\partial \Psi}{\partial t} = \Psi_0 \, 2\pi\nu \cos 2\pi\nu t$$

Differentiating it further –

$$\frac{\partial^2 \Psi}{\partial t^2} = -\Psi_0 \, 4\pi^2 \nu^2 \sin 2\pi\nu t$$

$$\boxed{\therefore \frac{\partial^2 \Psi}{\partial t^2} = -4\pi^2 \nu^2 \Psi} \quad (4)$$

$$\boxed{\lambda = \frac{h}{mv}} \quad (1)$$

$$\boxed{\therefore \Psi = \Psi_0 \sin 2\pi\nu t} \quad (2)$$

$$\boxed{\frac{\partial^2 \Psi}{\partial t^2} = v^2 \frac{\partial^2 \Psi}{\partial x^2}} \quad (3)$$

$$\boxed{\therefore \frac{\partial^2 \Psi}{\partial t^2} = -4\pi^2 \nu^2 \Psi} \quad (4)$$

As, $\text{frequency } (\nu) = \dfrac{\text{velocity } (v)}{\text{wavelength } (\lambda)}$

Equation 4 Becomes-

$$\frac{\partial^2 \Psi}{\partial t^2} = -\frac{4\pi^2 v^2}{\lambda^2} \Psi \quad (5)$$

From equation 3 and 5 we get

$$v^2 \frac{\partial^2 \Psi}{\partial x^2} = -\frac{4\pi^2 v^2}{\lambda^2} \Psi$$

$$\boxed{\frac{\partial^2 \Psi}{\partial x^2} + \frac{4\pi^2}{\lambda^2} \Psi = 0} \quad (6)$$

$$\boxed{\lambda = \frac{h}{mv}} \quad (1)$$

$$\boxed{\frac{\partial^2 \Psi}{\partial x^2} + \frac{4\pi^2}{\lambda^2}\Psi = 0} \quad (6)$$

$$\boxed{\frac{\partial^2 \Psi}{\partial x^2} + \frac{4\pi^2 m^2 v^2}{h^2}\Psi = 0} \quad (7)$$

Energy $E = K.E. + P.E.$

$$\therefore E = \frac{1}{2} mv^2 + V$$

$$\boxed{\therefore m^2 v^2 = 2m\,(E - V)} \quad (8)$$

Substituting 8 in 7 we get

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{4\pi^2}{h^2} 2m(E - V)\,\Psi = 0$$

$$\boxed{\frac{\partial^2 \Psi}{\partial x^2} + \frac{2m}{\hbar^2}(E - V)\,\Psi = 0} \qquad \because \hbar = \frac{h}{2\pi}$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{\partial^2 \Psi}{\partial x^2} + V\,\Psi = E\,\Psi} \quad (9)$$

This is Schrodinger's Time Independent Wave Equation (STIE)

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$$\boxed{\frac{\partial^2\Psi}{\partial x^2} + \frac{2m}{\hbar^2}(E - V)\,\Psi = 0}$$

$$\boxed{-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\,\Psi = E\,\Psi}$$

- This equation is independent of time and gives a steady value.
- It is particularly useful when the energy of the particle is very small as compared to its rest energy.
- In most atomic problems, energy of the particle is very small when compared to rest energy.

Schrodinger's Time Dependent Wave Equation (STDE)

- Let us consider a free particle of mass 'm' moving with velocity 'v' in one dimension.
- Let 'p' be the momentum and 'E' be the energy of the particle.
- By the term free particle, it means that no forces are acting on it and its total energy E is entirely kinetic energy.

$$\boxed{\text{Kinetic Energy} = \frac{1}{2}mv^2 = \frac{1}{2m}m^2v^2 = \frac{p^2}{2m}} \qquad (1)$$

This moving particle is associated with De Broglie waves which have wavelength λ and frequency ν. These are related as -

$$\boxed{p = \frac{h}{\lambda} = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k} \qquad (2)$$

$k = \frac{2\pi}{\lambda}$ is propagation constant

$$\boxed{E = h\nu = \frac{h}{2\pi} \times 2\pi\nu = \hbar\omega} \qquad (3)$$

$\omega = 2\pi\nu$ is angular frequency

Schrodinger's Time Dependent Wave Equation (STDE)

$$E = \frac{p^2}{2m}$$ (1)

$$p = \frac{h}{\lambda} = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k$$ (2)

$$E = h\nu = \frac{h}{2\pi} \times 2\pi\nu = \hbar\omega$$ (3)

From equation (1) (2) and (3) we get

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}$$ (4)

The wave equation describing the travelling waves must satisfy equation (4)

The wave function associated with the particle may by a sine, cosine or exponential function of $(kx - \omega t)$. Such functions are harmonic and can be superimposed, thereby giving a wave packet.

Let, $\Psi = \Psi_0 \, e^{i(kx-\omega t)}$ (5)

Schrodinger's Time Dependent Wave Equation (STDE)

$$\boxed{p = \frac{h}{\lambda} = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k} \quad (2)$$

$$\boxed{E = h\nu = \frac{h}{2\pi} \times 2\pi\nu = \hbar\omega} \quad (3)$$

$$\boxed{\hbar\omega = \frac{\hbar^2 k^2}{2m}} \quad (4)$$

$$\boxed{\text{Let,} \quad \Psi = \Psi_0\, e^{i(kx-\omega t)}} \quad (5)$$

Let us consider

$$i\hbar \frac{\partial \Psi}{\partial t} = i\hbar \frac{\partial\left(\Psi_0\, e^{i(kx-\omega t)}\right)}{\partial t}$$

$$\therefore i\hbar \frac{\partial \Psi}{\partial t} = i\hbar\,(-i\omega)\,\Psi_0\, e^{i(kx-\omega t)}$$

$$\boxed{\therefore i\hbar \frac{\partial \Psi}{\partial t} = \hbar\omega\Psi = E\,\Psi} \quad (6)$$

Schrodinger's Time Dependent Wave Equation (STDE)

$$p = \frac{h}{\lambda} = \frac{h}{2\pi} \times \frac{2\pi}{\lambda} = \hbar k \qquad (2)$$

$$E = h\nu = \frac{h}{2\pi} \times 2\pi\nu = \hbar\omega \qquad (3)$$

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} \qquad (4)$$

Let, $\Psi = \Psi_0 \, e^{i(kx-\omega t)}$ (5)

$$\therefore i\hbar \frac{\partial \Psi}{\partial t} = \hbar\omega\Psi = E\,\Psi \qquad (6)$$

Let us consider

$$\frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = \frac{-\hbar^2}{2m} \frac{\partial^2 \left(\Psi_0 \, e^{i(kx-\omega t)}\right)}{\partial^2 x}$$

$$\therefore \frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = \frac{-\hbar^2}{2m} (ik)^2 \Psi_0 \, e^{i(kx-\omega t)}$$

$$\therefore \frac{-\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} = \frac{\hbar^2 k^2}{2m} \Psi = \frac{p^2}{2m} \Psi \qquad (7)$$

From equation (6) and (7) we get

$$E = i\hbar \frac{\partial}{\partial t} \qquad p = i\hbar \frac{\partial}{\partial x} \qquad (8)$$

Schrodinger's Time Dependent Wave Equation (STDE)

$$p = \hbar k \quad (2)$$

$$E = \hbar\omega \quad (3)$$

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} \quad (4)$$

Let, $\Psi = \Psi_0\, e^{i(kx-\omega t)}$ (5)

$$\therefore i\hbar\frac{\partial\Psi}{\partial t} = \hbar\omega\Psi = E\,\Psi \quad (6)$$

$$E = i\hbar\frac{\partial}{\partial t} \qquad p = i\hbar\frac{\partial}{\partial x}$$

$$\therefore \frac{p^2}{2m} = \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} \quad (8)$$

If the particle is not free particle, the total energy of the particle is -

$$E = \frac{p^2}{2m} + V$$

Let us write these as operators operating on wave function Ψ

$$\therefore E\,\Psi = \frac{p^2}{2m}\Psi + V\,\Psi$$

Using 8

$$i\hbar\frac{\partial\Psi}{\partial t} = \frac{-\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\,\Psi$$

This is Schrodinger's Time Dependent Wave Equation (STDE)

Schrodinger's Time Dependent Wave Equation (STDE)

$$\boxed{i\hbar\frac{\partial\Psi}{\partial t} = \frac{-\hbar^2}{2m}\ \frac{\partial^2\Psi}{\partial x^2} + V\,\Psi}$$

- The solution of time dependent Schrodinger equation results in states which are not stationary.
- The corresponding probability densities are not constant and they vary with time.
- Let free electron in a box. If we apply an electric field on the box periodically, the original wave functions and the corresponding probability densities of the electron are not constant and they do depend on the time.

STIE

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\Psi = E\Psi$$

The solution of time independent Schrodinger equation results in stationary states, where the probability density is independent of time.

STDE

$$i\hbar\frac{\partial\Psi}{\partial t} = \frac{-\hbar^2}{2m}\;\frac{\partial^2\Psi}{\partial x^2} + V\Psi$$

The solution of time dependent Schrodinger equation results in states which are not stationary. The corresponding probability densities vary with time.

As Newton's laws predict the future behavior of a dynamic system in classical mechanics, Schrodinger's equations are used to predict future behavior in quantum mechanics.